1.5.1 Hydrogen ion concentration and pH

The acidic or basic nature of a solution is measured by H^+ ion concentration. The conventional units such as moles/l or g/l are not commonly used to express H^+ ion concentration. Sorenson (1909) introduced the term pH to express H^+ ion concentration. pH is defined as the negative logarithm of H^+ ion concentration.

$$pH = - \log [H^+]$$

DO YOU KNOW?

pH is an abbreviation for "power of hydrogen" where "p" represents the German word 'potenz' for power, and H is the element symbol for hydrogen. In short, pH stands for "potential of hydrogen". It was described by Soren Peder Lauritz Sorensen in 1909.

pH Scale

The pH is a narrow scale, ranging from 0 to 14 which corresponds to 1 M solution to 10^{-14} M solution of $[H^+]$. Pure water has an equal concentration of H^+ and OH^- ions i.e. 10^{-7} M each. Thus, pure water has a pH 7 which is neutral. Solutions with pH less than 7 are said to be acidic while those with pH greater than 7 are alkaline. lt must be remembered that the term acidic or alkaline are not absolute but only relative. Thus, a solution with pH 3.0 is more acidic when compared with a solution of pH 4.5. A rise in H^+ concentration decreases pH while a fall in H^+ concentration increases pH. The reverse is true for OH^- concentration. The pH of a solution containing 1N $[H^+]$ is 0 while that containing 1N $[OH^-]$ is 14.

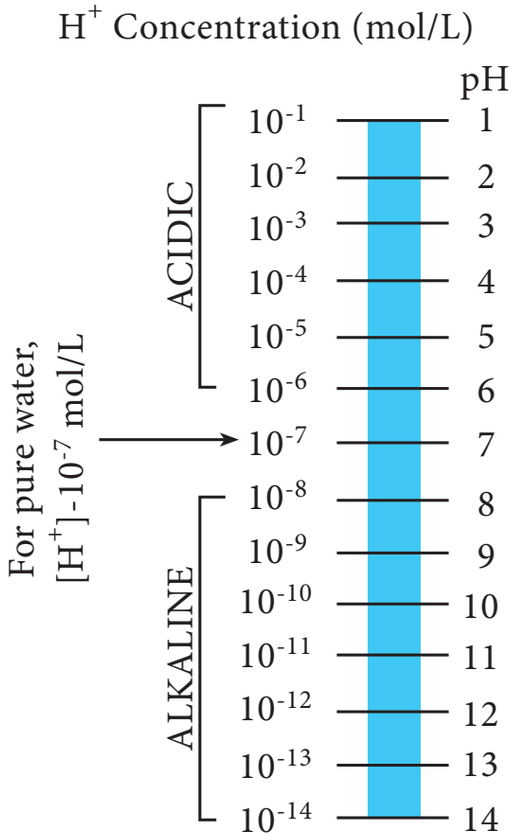


Figure 1. pH scale

Evaluate 1

Calculate the H^+ ion concentration if pH = 8.5

1.5.2 Buffers

A buffer is an aqueous solution consisting of a mixture of a weak acid and its conjugate base or a weak base and its conjugate acid. A buffer resists changes in pH when a small amount of strong acid or base is added and hence it maintains the pH of a solution. Many organisms thrive only in a relatively small pH range so they utilize buffer systems to maintain a constant pH. Buffers are more effective within 1.0 pH unit range of its pK_a value.

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